The documentation and process conversion measures necessary to comply with this revision shall be completed by 22 July 1994

INCH-POUND

MIL-S-19500/182D
22 April 1994
SUPERSEDING
MIL-S-19500/182C
6 February 1967

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES 2N720A, 2N1893, 2N1893S, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (similar to TO-18 and TO-5).

1.3 Maximum ratings.

Type	P_T 1/ T_C = +25°C	P_T 2/ T_A = +25°C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	V_{CER} R_{BE} = 10 Ω	T_J and T_{STG}
	W	W	V dc	V dc	V dc	mA dc	V dc	°C
2N720A	1.8	0.5	120	7	80	500	100	-65 to +200
2N1893	3.0	0.8	120	7	80	500	100	-65 to +200
2N1893S	3.0	0.8	120	7	80	500	100	-65 to +200

1/ Derate linearly at 10.3 mW/°C for type 2N720A and 17.2 mW/°C for type 2N1893 and 2N1893S for T_C > +25°C.
2/ Derate linearly at 2.86 mW/°C for type 2N720A and 4.57 mW/°C for type 2N1893 and 2N1893S for T_A > +25°C.

1.4 Primary electrical characteristics.

Limits	h_{FE1} 1/	h_{FE2} 1/	h_{FE3} 1/	|hfe|	$V_{CE(SAT)}$
	V_{CE} = 10 V dc I_C = 0.1 mA dc	V_{CE} = 10 V dc I_C = 10 mA dc	V_{CE} = 10 V dc I_C = 150 mA dc	f = 20 MHz V_{CE} = 10 V dc I_C = 50 mA dc	I_C = 50 mA dc I_B = 5.0 mA dc
					V dc
Min	20	35	40	3.0	0.2
Max			120	10	1.2

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defence Electronics Supply Center ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 9 (similar to TO-18), type T-2A for 2N720A, figure 8 (similar to TO-5), type T-1A for 2N1893, and T-1C for 2N1893S.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and table II herein.

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4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels only
1/	Thermal impedance (see 4.3.2)
9	Not applicable
11	h_{FE3}; I_{CBO1}
12	See 4.3.1
13	ΔI_{CBO1} = 100 percent of initial value or 5 nA dc; whichever is greater; Δh_{FE3} = ±15 percent of initial value; subgroup 2 of table I herein.

1/ Thermal impedance shall be performed anytime after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750.
V_{CB} = 60 V dc.
2N720A, P_T = 500 mW dc.
2N1893, 2N1893S, P_T = 800 mW dc.

NOTE: No heat sink or forced air cooling of the devices shall be permitted.

4.3.2 Thermal impedance $Z_{\theta JC}$ measurements for screening. The $Z_{\theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3131. The maximum limit and conditions for $Z_{\theta JC}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not to exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot.

4.3.2.1 Thermal impedance ($Z_{\theta JC}$ measurements) for initial qualification or requalification. The $Z_{\theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3131 (read and record value $Z_{\theta JC}$) derived conditions, limits, and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JC}$ in group A subgroup 2 inspection:

a. I_M measuring current: 10 mA.
b. I_H forward heating current: 10 A.
c. t_H heating time: 10 ms.
d. t_{MD} measurement delay time: 100 µs maximum.

The maximum limit for $Z_{\theta JC}$ in group A, subgroup 2 is $Z_{\theta JC}$ (maximum) = 4.5°C/W.

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Inspection Method Examination

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end- points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750. V_{CB} = 60 V dc. 2N720A, P_T = 500 mW dc; 2N1893, 2N1893S, P_T = 800 mW dc.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Lead fatigue: Test condition E.
C6	1026	T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750. V_{CB} = 60 V dc; P_T = 500 mW for 2N720A, P_T = 800 mW for 2N1893 and 2N1893S.

4.4.4 Group E inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
E1	1051	Condition C, 500 cycles; 22 devices, c = 0.
E1	1071	
E2	1038	80 to 85% of rated V_R, T_A = 150°C, t = 1,000 hours
E3	----	Not Applicable.
E4	3131	See 4.5.2, 22 devices, c = 0.
E5	----	Not Applicable.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance (qualification only). Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 28 mA dc (2N720A), and 47 mA dc (2N1893 and 2N1893S).

b. Collector to emitter voltage magnitude ≥ 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C ≤ T_R ≤ 75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = (2N720A) 97°C/W, and (2N1893 and 2N1893S) 58°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.4.1	$Z_{\theta JC}$		4.5	°C/W
Breakdown voltage collector to base	3001	Bias condition D, I_C = 100 µA dc, pulsed (see 4.5.1)	$V_{(BR)CBO}$	120		V dc
Breakdown voltage emitter to base	3026	Bias condition D, I_E = 100 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	7		V dc
Breakdown voltage collector-emitter	3011	Bias condition D, I_C = 30 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Breakdown voltage collector-emitter	3011	Bias condition D, I_C = 100 mA dc, pulsed (see 4.5.1), R_{BE} = 10 ohms	$V_{(BR)CER}$	100		V dc
Collector to base cutoff current	3036	Bias condition D V_{CB} = 90 V dc	I_{CBO1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5 V dc	I_{EBO}		10	nA dc
Collector to emitter saturated voltage	3071	I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$	.3	5.0	V dc
Base emitter saturated voltage	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.3	V dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA dc, pulsed (see 4.5.1)	h_{FE1}	20		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE2}	35		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE3}	40	120	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE4}	20		

See footnotes at end of table.

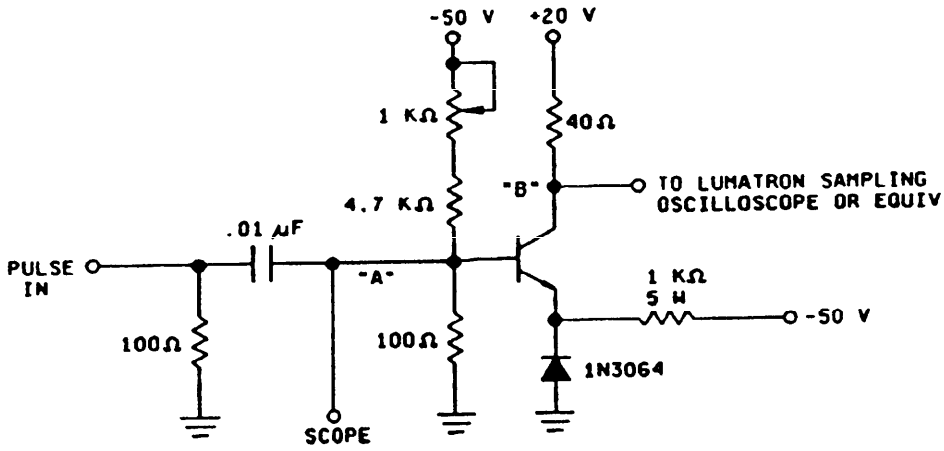
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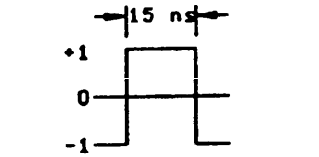
TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 90 V dc	I_{CBO2}		15	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE1}	20		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc, I_C = 1 mA dc.	h_{fe}	35	100	
		V_{CE} = 10 V dc, f = 1 kHz, I_C = 5 mA dc	h_{fe}	45		
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 50 mA dc, f = 20 MHz	$\|h_{fe}\|$	3	10	
Small-signal short-circuit input impedance	3201	V_{CB} = 10 V dc, I_C = 5 mA dc,	h_{ib}	4	8	ohms
Small-signal open-circuit output admittance	3216	V_{CB} = 10 V dc, I_C = 5 mA dc,	h_{ob}	0	0.5	µohms
Small-signal open-circuit reverse voltage-transfer ratio	3211	V_{CB} = 10 V dc, I_C = 5 mA dc,	h_{rb}		1.5×10^{-4}	
Open-circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}	5	15	pF
Pulse response	3251	Test condition A, except test circuit and pulse requirements See figure 1 herein.	t_{on} + t_{off}		30	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.
2/ Need not be performed in screen 13.

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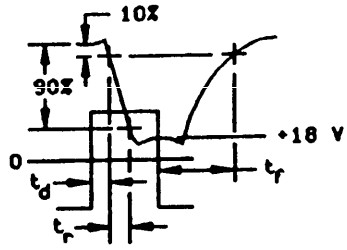




WAVEFORM AT "A"
FROM LUMATRON Hg PULSE
GENERATOR OR EQUIVALENT



WAVEFORM AT "B"



NOTES:
1. The rise time (T_r) of the applied pulse shall be 2.0 ≤ nanoseconds, duty cycle ≤ 2%, and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 nsec.

FIGURE 1. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 14, 19, 85

Preparing activity:
DLA - ES

(Project 5961-1632)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/182D	2. DOCUMENT DATE (YYMMDD) 94/04/22

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N720A, 2N1893, 2N1893S, JAN, JANTX, and JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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